# Do Now · retrieval

## Lesson 106 · What an Operating System Does

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Distinguish RAM from virtual storage.

### Q2

In a BST, where is a value less than the current node placed?

### Q3

Why can an array provide direct indexed access?

### Q4

A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Manage processes — The OS creates and schedules processes; programs share CPU time.

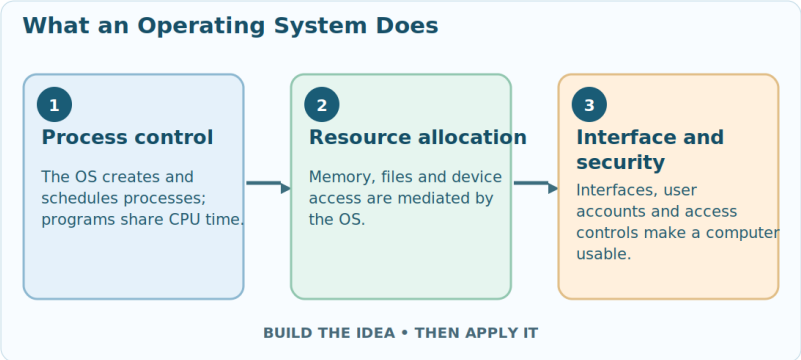
## Core idea 2

Manage resources — Memory, files and device access are mediated by the OS.

## Core idea 3

Provide services — Interfaces, user accounts and access controls make a computer usable.

**Key visual · complete concept**



# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

operating system — System software managing hardware resources and services.

process — An executing instance of a program.

device driver — Software allowing the OS to communicate with hardware.

spooling — Queuing data such as print jobs for a slower peripheral.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

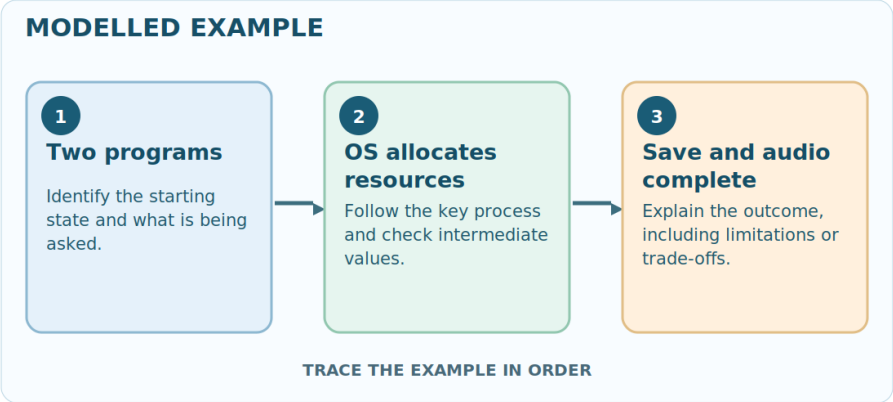
## Problem

A word processor saves a document while music is playing. Explain three OS services involved.

## Follow the worked method

1. Model answer — The OS allocates CPU time to both processes; the file system writes the document to storage; a device driver controls sound output. Memory management keeps process data protected. — Check: link each claim to the scenario and use precise OCR terminology.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer every question in BLACK PEN. Show steps and use OCR command words.

## Question 1

Describe three functions of an operating system, linking each to a user action. [6]

## Question 2

A computer has three applications requesting the printer. Explain how the OS can handle this. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer every question in BLACK PEN. Show steps and use OCR command words.

## Question 3

Evaluate the claim that application programs can directly manage all hardware without system software. [8]

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

The operating system only provides a graphical user interface.

## Use key vocabulary from memory

1. Explain operating system in your own words.

2. Explain process in your own words.

3. Explain device driver in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.